

ECE 280 Lab 3 – TA Solution and Troubleshooting Manual

Signals and Systems: CT/DT System Properties Explorer

Use this document as a fast grading and troubleshooting guide for:

- Lab 3 manual and report template submissions
- System-property verification from hardware-acquired data
- TA checkoffs during in-lab demonstrations

1 Quick Pass/Flag Menu

Pass Immediately If All Are True

- Student presents physical evidence for all three circuits (RC, diode clipper, comparator).
- All required screenshots include student name/ID and are legible.
- MATLAB verification uses measured input/output data (not synthetic-only arrays).
- Superposition and TI tests include clear overlays and reported NRMSE values.
- Final truth table is evidence-backed and consistent with measured behavior.

Flag for Review If Any Are True

- Property claims are made without scope evidence and/or numerical checks.
- No unique test inputs (students reuse common canned waveforms without assigned seed variation).
- Data appears copied across circuits (identical traces for RC/clipper/comparator).
- MATLAB script cannot identify hardware data file origin.

Table 1: Reference Truth Table (Expected in Most Correct Builds)

System	Linear	Time-Invariant	Causal	BIBO Stable
RC low-pass filter	Yes	Yes	Yes	Yes
Diode clipper	No	Yes	Yes	Yes
Comparator (fixed threshold)	No	Yes	Yes	Yes

2 Instructor Key: Expected Property Outcomes

Assume ideal operating conditions and correctly built circuits.

Notes for TAs:

- Clipper and comparator are memoryless nonlinear maps, so they generally remain TI but violate superposition.
- Comparator with drifting threshold or supply instability may appear weakly non-TI in practice; this should be discussed, not ignored.

3 Pre-Lab Solution Guidance

Required Conceptual Targets

- Correct formal definitions of linearity, TI, causality, and BIBO stability.
- Predicted outcomes for all three circuits with mechanism-based justification.
- Test design includes two distinct inputs, nontrivial coefficients a, b , and a measurable time shift Δ .

Accept/Revise Criteria

- Accept if definitions are mathematically valid and predictions are physically coherent.
- Revise if student confuses “bounded output” with “low noise” or treats TI as “periodic output.”

4 Measurement Targets and Acceptance Bands

RC Filter Baseline

- Expect attenuation and phase lag dependent on frequency relative to cutoff.
- Accept if gain/phase trends are physically correct even with moderate component tolerance.

Diode Clipper Baseline

- Clipping threshold typically near diode conduction region (depends on topology/bias).
- Accept if waveform clearly shows limited peaks and threshold annotation is reasonable.

Comparator Baseline

- Output should switch between rails (or logic levels) around threshold crossing.
- Accept if transition behavior is consistent with threshold and input crossing events.

5 Numerical Verification Key (MATLAB)

Linearity Test Expectations

Students test:

$$\mathcal{T}\{ax_1 + bx_2\} \quad \text{vs} \quad a\mathcal{T}\{x_1\} + b\mathcal{T}\{x_2\}$$

Expected trends:

- RC filter: low residual, small NRMSE (often $< 10\%$ with good alignment).
- Diode clipper: clear failure where clipping activates; moderate/high NRMSE expected.
- Comparator: strong failure (binary nonlinearity), usually high NRMSE except trivial small-signal regimes.

Time-Invariance Test Expectations

Students test:

$$\mathcal{T}\{x(t - \Delta)\} \quad \text{vs} \quad y(t - \Delta)$$

Expected trends:

- All three systems should be TI in a stable setup with fixed parameters.
- Elevated TI NRMSE often indicates poor synchronization, trigger drift, or threshold drift rather than true TI failure.

6 TA Checkoff Script (2–3 Minutes per Team)

1. Ask team to state predicted truth table before showing data.
2. Verify one scope capture per circuit with visible settings and student ID.

3. Ask for one MATLAB linearity overlay and one TI overlay (any circuit).
4. Ask: “Why did this property pass/fail physically?”
5. Confirm final report will include handwritten truth table and code/data linkage.

7 Troubleshooting Matrix

Table 2: Common Failure Modes and Corrective Actions

Observed Symptom	Likely Cause	TA Action
No clipping visible in clipper output	Diode reversed, incorrect reference node, or amplitude too low	Check diode orientation and threshold path; increase input amplitude above expected clipping level.
Comparator output stuck high/low	Threshold out of range, missing ground reference, wiring error	Verify common ground and reference divider; sweep threshold and confirm crossing events.
RC output looks identical to input at all frequencies	Cutoff too high or probe connected before RC node	Recheck node location and component values; test with higher input frequency to reveal filtering.
TI test fails for all circuits	Time alignment/trigger mismatch, not intrinsic circuit behavior	Enforce common trigger method and post-alignment in MATLAB before property claim.
Linearity appears to pass clipper/comparator	Inputs too small to engage non-linearity	Increase amplitude to activate clipping/switching region and repeat superposition test.
MATLAB NRMSE unstable/run-to-run variation	Inconsistent sample window, serial timing jitter, poor preprocessing	Use fixed acquisition length, consistent trigger point, detrend/align before computing metrics.

8 Grading Menu Aligned to Lab 3 Rubric (100 pts)

- Pre-lab quality (15): definitions, predictions, and test design are rigorous.
- Circuit implementation/evidence (20): all three circuits built and documented.
- Linearity verification (15): valid superposition workflow and justified conclusion.
- TI verification (15): proper shifted-input methodology and justified conclusion.
- Causality/stability evaluation (15): bounded tests and anticipatory behavior check.
- Truth table/technical reasoning (10): final table is coherent and evidence-backed.
- Code/data authenticity (10): MATLAB clearly uses measured hardware data.

Suggested Deductions

- Missing ID on required evidence: -3 to -10 , depending on extent.
- No quantitative metric (NRMSE or equivalent): -4 to -8 .
- Property conclusions unsupported by data: -5 to -12 .
- Synthetic-only MATLAB verification: -8 to -15 .

9 Minimum Passing Submission Standard

A minimally acceptable submission should include:

- Physical evidence of all three circuits.
- At least one valid linearity and one valid TI test workflow.
- Final truth table with short justifications per system.
- MATLAB script excerpts tied to acquired datasets.
- Student identification visible in required screenshots.